# UNIVERSITY OF MYSORE

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## MYSORE UNIVERSITY SCHOOL OF ENGINEERING

Manasagangotri campus, Mysuru-570006
(Approved by AICTE, New Delhi)

**A Project Phase-II (21CDP81) Report**

**On**

# "AI POWERED PDF QUERY SYSTEM"

Submitted in partial fulfilment for the award of the degree of
Bachelor of Engineering
in
Computer Science and Design


**Submitted By**

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**DEPARTMENT OF COMPUTER SCIENCE AND DESIGN**

# CERTIFICATE

This is to certify that the **Project Phase-II (21CDP81)** title entitled as **"AI Powered PDF Query System"** is a bonafide work carried out by **Niharika B , Srihari S , Utsav M , Manju Samartha K S, VIII Semester** bearing **Register No. 21SECD26 , 21SECD45 , 21SECD52, 21SECD92,** are students of **Computer Science and Design, in** partial fulfillment for the award of **Bachelor of Engineering** in the **Department of Computer Science and Design, Mysore University School of Engineering, University of Mysore, Mysuru**. It is certified that all corrections/suggestions indicated for have been executed by the above-mentioned candidate.

Signature of the Head of the Department
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Signature of Faculty Incharge,
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Signature of the Director
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# DECLARATION

**We**, **Niharika B, Srihari S, Utsav M, Manju Samartha K S,** bearing the Register No.**21SECD26, 21SECD45, 21SECD52, 21SECD92,** student of VIII semester, **Bachelor of Engineering** in the **Department of Computer Science and Design, University of Mysore, Mysuru**. Declare that the **Project Phase-II (21CDP81) entitled on "AI Powered PDF Query System**", has been duly executed by us, under faculty incharge **Poornashree Narayani S Kulkarni.**

The Project Phase-II report of the above entitled is submitted by us in the view of partial fulfillment of the requirement for the award of a Bachelor of Engineering degree in Computer Science and Design by the Department of Computer Science and Design, Mysore University School of Engineering, University of Mysore, Mysuru. During the academic year **2024-25** further, the matter embodied in the report has not been submitted previously by anybody for the award of any degree/diploma/certificate programme.

**Date:**15/07/2025
**Place:** Mysuru

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**Niharika B**
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**Manju Samartha K S**

# ABSTRACT

In recent years, the need for intelligent systems that can interpret and summarize document-based information has grown significantly. This project presents the design and implementation of an AI-powered web application that allows users to interact with PDF files through a conversational interface. The system enables users to upload PDF documents, ask questions in natural language, and receive accurate responses based on the document content mimicking the capabilities of modern AI assistants like ChatGPT.

The solution adopts a modular architecture where the backend, powered by Hugging Face's transformer models, is hosted on a separate machine. This backend is responsible for parsing PDF documents, performing semantic chunking, embedding generation, and producing context-aware responses using an LLM (Large Language Model). The frontend is developed using React.js, Tailwind CSS, and Framer Motion to deliver a sleek, animated, and responsive user interface. It supports drag-and-drop PDF upload, animated typing effects, and chat-style responses to enhance user engagement.

The project leverages modern AI-driven tools like Vercel v0 for rapid UI prototyping and integrates best practices in both frontend design and backend model orchestration. The system demonstrates efficient document understanding, smooth user interaction, and scalability for larger datasets. Future enhancements include OCR support, multilingual capabilities, and deployment on cloud infrastructure for broader accessibility. This work highlights the synergy between NLP, user-centric design, and modular development in building intelligent document-interaction platforms.

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